

Practice Questions for 2024-10-03-DualSimplexMatrix

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1 Questions

1. Consider the primal problem (P) in Slide 2. The dual simplex method is employed because (P) is not primal feasible. What specific characteristic of the primal problem necessitates the use of the dual simplex method, and why is the dual problem (D) initially feasible?
 - a. The objective function of (P) has all positive coefficients.
 - b. The right-hand side of at least one constraint in (P) is negative.
 - c. The dual problem (D) has an unbounded objective function.
 - d. The constraints in (P) are inconsistent.
 - e. The objective function of (P) is unbounded.
2. In Slide 3, the primal and dual dictionaries are presented. Explain how the dual dictionary is derived from the primal dictionary, and why the origin is dual-feasible but not primal-feasible.
 - a. The dual dictionary is obtained by negating the primal dictionary; the origin is dual-feasible because all variables are non-negative.
 - b. The dual dictionary is obtained by transposing the coefficient matrix of the primal dictionary and changing the signs of the constants; the origin is dual-feasible because the constants in the dual constraints are non-negative.
 - c. The dual dictionary is obtained by inverting the coefficient matrix of the primal dictionary; the origin is dual-feasible because the objective function is minimized.
 - d. The dual dictionary is identical to the primal dictionary; the origin is dual-feasible because the dual problem is always feasible.
 - e. The dual dictionary is obtained by taking the transpose of the primal dictionary and changing the signs of the objective function coefficients; the origin is dual-feasible because the dual problem is always feasible.
3. In Slide 4, the dual simplex method iteratively transforms the primal and dual dictionaries. Explain the relationship between the primal and dual dictionaries at each step, and how the process terminates.
 - a. The primal and dual dictionaries are independent; the process terminates when the primal objective function is maximized.
 - b. The dual dictionary is the negative transpose of the primal dictionary; the process terminates when all right-hand sides in the primal dictionary are non-negative.
 - c. The primal and dual dictionaries are identical; the process terminates when the dual objective function is minimized.

- d. The dual dictionary is obtained by inverting the primal dictionary; the process terminates when the primal objective function is minimized.
 - e. The primal and dual dictionaries are unrelated; the process terminates when both the primal and dual problems are feasible.
4. Slides 5 and 6 introduce a "Dual Based Phase I Algorithm." Explain why a Phase I algorithm is necessary in this case, and how the modification in Slide 6 achieves dual feasibility.
- a. A Phase I algorithm is needed because the primal problem is unbounded; the modification in Slide 6 changes the objective function to ensure primal feasibility.
 - b. A Phase I algorithm is needed because both the primal and dual problems are infeasible at the origin; the modification in Slide 6 changes the objective function to make the dual problem feasible at the origin.
 - c. A Phase I algorithm is needed because the dual problem is unbounded; the modification in Slide 6 changes the constraints to ensure dual feasibility.
 - d. A Phase I algorithm is unnecessary; the modification in Slide 6 is simply a simplification of the problem.
 - e. A Phase I algorithm is needed because the primal problem is infeasible; the modification in Slide 6 changes the constraints to make the dual problem feasible.
5. Consider the primal problem (P) in Slide 2. The dual simplex method is employed because (P) is not primal feasible. What specific characteristic of the primal problem necessitates the use of the dual simplex method, and why is the dual problem (D) initially feasible?
- a. The primal problem (P) has negative entries in the right-hand side vector b , making it infeasible at the origin. The dual problem (D) is initially feasible because the origin satisfies its constraints.
 - b. The primal problem (P) has an unbounded objective function. The dual problem (D) is initially feasible because it is a minimization problem.
 - c. The primal problem (P) has redundant constraints. The dual problem (D) is initially feasible because it has fewer constraints.
 - d. The primal problem (P) has an inconsistent constraint set. The dual problem (D) is initially feasible because it is always feasible.
 - e. The primal problem (P) has a non-linear objective function. The dual problem (D) is initially feasible because it is a linear problem.
6. In the dual simplex method, why is the dual problem initially feasible even if the primal problem is not?
- a. The dual problem's constraints are always feasible regardless of the primal problem's feasibility.
 - b. The dual problem's feasibility is independent of the primal problem's feasibility.
 - c. The dual problem's objective function is always minimized at the origin.
 - d. The dual problem's constraints are derived from the primal problem's objective function, which ensures dual feasibility.
 - e. The dual problem's constraints are the negative transpose of the primal problem's constraints, and the non-negativity of the dual variables ensures dual feasibility at the origin if the primal objective function coefficients are all non-positive.

7. In Slide 3, the origin is described as dual-feasible but not primal-feasible. Explain why this is the case, referencing the primal and dual dictionaries.
 - a. The primal dictionary has positive right-hand sides, while the dual dictionary has negative right-hand sides.
 - b. The primal dictionary has negative right-hand sides in some constraints, indicating primal infeasibility, while the dual dictionary has non-negative right-hand sides, indicating dual feasibility.
 - c. The primal dictionary shows that the objective function is unbounded, while the dual dictionary shows a bounded objective function.
 - d. The primal dictionary has all non-negative coefficients, while the dual dictionary has some negative coefficients.
 - e. The primal dictionary has a negative objective function value, while the dual dictionary has a positive objective function value.
8. In Slide 4, the dual simplex method iteratively transforms the primal and dual dictionaries. Describe the key transformation that occurs in each step, and explain how this relates to maintaining dual feasibility.
 - a. Each step involves pivoting on a negative entry in the primal dictionary, maintaining primal feasibility.
 - b. Each step involves pivoting on a positive entry in the dual dictionary, maintaining dual feasibility.
 - c. Each step involves pivoting on a negative entry in the dual dictionary, maintaining dual feasibility.
 - d. Each step involves pivoting on a positive entry in the primal dictionary, maintaining primal feasibility.
 - e. Each step involves simultaneously pivoting on entries in both the primal and dual dictionaries, maintaining both primal and dual feasibility.
9. Slides 5 and 6 introduce a "Dual Based Phase I Algorithm." Explain why a Phase I algorithm is necessary in this case, and how the modification in Slide 6 achieves dual feasibility.
 - a. A Phase I algorithm is needed because the primal problem is unbounded; Slide 6 modifies the objective function to ensure boundedness.
 - b. A Phase I algorithm is needed because the dual problem is infeasible; Slide 6 modifies the constraints to ensure dual feasibility.
 - c. A Phase I algorithm is needed because both the primal and dual problems are infeasible at the origin; Slide 6 modifies the objective function to achieve dual feasibility.
 - d. A Phase I algorithm is needed because the primal problem is infeasible; Slide 6 modifies the constraints to ensure primal feasibility.
 - e. A Phase I algorithm is needed because the dual problem is unbounded; Slide 6 modifies the objective function to ensure boundedness.
10. Consider the primal problem (P) in Slide 2. The dual simplex method is employed because (P) is not primal feasible. What specific characteristic of the primal problem necessitates the use of the dual simplex method, and why is the dual problem (D) initially feasible?
 - a. The primal problem (P) has negative entries in the right-hand side vector b , making it infeasible at the origin. The dual problem (D) is initially feasible because the origin satisfies its constraints.

- b. The primal problem (P) has an unbounded objective function. The dual problem (D) is initially feasible because it is a minimization problem.
- c. The primal problem (P) has redundant constraints. The dual problem (D) is initially feasible because it has fewer constraints.
- d. The primal problem (P) has an inconsistent set of constraints. The dual problem (D) is initially feasible because it is always feasible.
- e. The primal problem (P) has a non-linear objective function. The dual problem (D) is initially feasible because it is a linear problem.

2 Answers

1. Consider the primal problem (P) in Slide 2. The dual simplex method is employed because (P) is not primal feasible. What specific characteristic of the primal problem necessitates the use of the dual simplex method, and why is the dual problem (D) initially feasible?
 - a. A positive objective function coefficient is not a condition for using the dual simplex method. The dual simplex method is used when the primal problem is infeasible.
 - b. The dual simplex method is used when the primal problem is infeasible, specifically when at least one constraint has a negative right-hand side. This is because the standard simplex method requires an initial feasible solution, which is not available in this case. The dual problem is initially feasible because the origin (all dual variables equal to zero) satisfies the dual constraints. The negative right-hand sides in the primal constraints translate to positive constants in the dual constraints, ensuring feasibility at the origin.
 - c. The dual problem's unboundedness is a consequence, not a cause, of primal infeasibility. The dual simplex method addresses primal infeasibility.
 - d. Inconsistent constraints in the primal problem would make both the primal and dual problems infeasible. The dual simplex method is used when the primal is infeasible but the dual is feasible.
 - e. An unbounded primal objective function would imply an infeasible dual problem, not a situation where the dual simplex method is applicable.
2. In Slide 3, the primal and dual dictionaries are presented. Explain how the dual dictionary is derived from the primal dictionary, and why the origin is dual-feasible but not primal-feasible.
 - a. Negating the primal dictionary does not produce the dual dictionary. The dual is obtained by a transpose and sign changes.
 - b. The dual dictionary is derived from the primal dictionary by taking the negative transpose of the coefficient matrix of the constraints and changing the signs of the constants. The origin is dual-feasible because the constants in the dual constraints (the right-hand sides of the z_i equations) are non-negative. The origin is not primal-feasible because the constants in the primal constraints (the right-hand sides of the w_i equations) are not all non-negative.
 - c. Inverting the coefficient matrix does not yield the dual dictionary. The dual is obtained by a transpose and sign changes.
 - d. The primal and dual dictionaries are not identical. The dual is obtained by a transpose and sign changes.
 - e. While the transpose is correct, changing the signs of the objective function coefficients is not part of the transformation. The constants in the constraints are what determine feasibility.
3. In Slide 4, the dual simplex method iteratively transforms the primal and dual dictionaries. Explain the relationship between the primal and dual dictionaries at each step, and how the process terminates.
 - a. The primal and dual dictionaries are closely related through the transformation described above. The process terminates when a primal feasible solution is found.
 - b. At each step, the dual dictionary is obtained by taking the negative transpose of the coefficient matrix of the primal constraints and changing the signs of the constants. The process terminates when all right-hand sides in the primal dictionary become non-negative, indicating that a primal feasible solution has been found. This also implies that the dual objective function has reached its minimum value.

- c. The primal and dual dictionaries are not identical. The dual is obtained by a transpose and sign changes. The process terminates when a primal feasible solution is found.
 - d. Inverting the primal dictionary does not yield the dual dictionary. The dual is obtained by a transpose and sign changes. The process terminates when a primal feasible solution is found.
 - e. The primal and dual dictionaries are closely related. The process terminates when a primal feasible solution is found, which implies that the dual objective function has reached its minimum.
4. Slides 5 and 6 introduce a "Dual Based Phase I Algorithm." Explain why a Phase I algorithm is necessary in this case, and how the modification in Slide 6 achieves dual feasibility.
- a. The primal problem's unboundedness is not the reason for a Phase I algorithm. The issue is infeasibility at the origin for both primal and dual problems.
 - b. A Phase I algorithm is necessary because both the primal and dual problems are infeasible at the origin (all variables equal to zero). The standard simplex method requires an initial feasible solution. The modification in Slide 6 changes the objective function of the primal problem (and consequently the dual objective function) to create a dual feasible problem at the origin. This allows the dual simplex method to proceed.
 - c. The dual problem's unboundedness is not the reason for a Phase I algorithm. The issue is infeasibility at the origin for both primal and dual problems.
 - d. A Phase I algorithm is crucial in this case to find a feasible starting point for the dual simplex method.
 - e. While the primal problem is infeasible, the modification in Slide 6 changes the objective function, not the constraints, to achieve dual feasibility.
5. Consider the primal problem (P) in Slide 2. The dual simplex method is employed because (P) is not primal feasible. What specific characteristic of the primal problem necessitates the use of the dual simplex method, and why is the dual problem (D) initially feasible?
- a. The correct answer is A. The dual simplex method is designed for situations where the primal problem is infeasible but the dual is feasible. The presence of negative entries in the right-hand side vector b of the primal problem indicates that the origin is not a feasible solution. The dual problem, however, is constructed in such a way that the origin often satisfies its constraints, making it initially feasible. This allows the dual simplex method to start its iterations from a feasible point in the dual space and iteratively move towards a feasible and optimal solution for the primal problem.
 - b. Incorrect. An unbounded primal problem implies an infeasible dual problem, not the other way around. The dual simplex method is not directly related to the unboundedness of the objective function.
 - c. Incorrect. Redundant constraints in the primal problem do not necessarily imply that the dual simplex method is needed. The dual simplex method addresses primal infeasibility, not redundancy.
 - d. Incorrect. An inconsistent constraint set in the primal problem would make both the primal and dual problems infeasible. The dual simplex method is used when the primal is infeasible but the dual is feasible.
 - e. Incorrect. The dual simplex method applies to linear programs. Non-linearity would require different solution techniques.

6. In the dual simplex method, why is the dual problem initially feasible even if the primal problem is not?
- a. **Incorrect.** The dual problem's feasibility is directly related to the primal problem's structure and coefficients.
 - b. **Incorrect.** The dual problem's feasibility is dependent on the primal problem's feasibility.
 - c. **Incorrect.** The dual problem's objective function is minimized, but not necessarily at the origin.
 - d. **Incorrect.** The dual problem's constraints are derived from the primal problem's constraints, not the objective function.
 - e. The dual problem's constraints are derived from the primal problem's constraints by taking the negative transpose. If the primal objective function coefficients are all non-positive, setting the dual variables to zero satisfies the dual constraints, making the origin dual-feasible. This is the case in the example presented in Slide 2, where the primal objective function has all negative coefficients.
7. In Slide 3, the origin is described as dual-feasible but not primal-feasible. Explain why this is the case, referencing the primal and dual dictionaries.
- a. **Incorrect.** The primal dictionary has negative right-hand sides, indicating infeasibility.
 - b. The primal dictionary in Slide 3 shows that the variables w_2 and w_3 have negative right-hand sides (-8 and -7 respectively), indicating that the origin is not a feasible solution for the primal problem. In contrast, the dual dictionary shows that the variables z_1 and z_2 have non-negative right-hand sides (1 and 1 respectively), indicating that the origin is a feasible solution for the dual problem.
 - c. **Incorrect.** The dictionaries do not indicate unboundedness in either problem at the origin.
 - d. **Incorrect.** The coefficients in the dictionaries do not determine feasibility at the origin.
 - e. **Incorrect.** The objective function values at the origin are not relevant to the feasibility of the origin.
8. In Slide 4, the dual simplex method iteratively transforms the primal and dual dictionaries. Describe the key transformation that occurs in each step, and explain how this relates to maintaining dual feasibility.
- a. **Incorrect.** Pivoting on a negative entry in the primal dictionary does not maintain primal feasibility; it's used to improve it.
 - b. **Incorrect.** Pivoting on a positive entry in the dual dictionary is not the core mechanism of the dual simplex method.
 - c. The dual simplex method maintains dual feasibility at each step. It selects a row with a negative right-hand side in the primal dictionary (indicating primal infeasibility) and then pivots on a negative entry in that row. This pivot operation simultaneously updates both the primal and dual dictionaries, ensuring that the dual dictionary remains feasible (all right-hand sides non-negative). The process continues until all right-hand sides in the primal dictionary are non-negative, indicating primal feasibility and optimality.
 - d. **Incorrect.** Pivoting on a positive entry in the primal dictionary is not the core mechanism of the dual simplex method.
 - e. **Incorrect.** The dual simplex method pivots on a single entry in the primal dictionary, which implicitly updates the dual dictionary to maintain dual feasibility.

9. Slides 5 and 6 introduce a "Dual Based Phase I Algorithm." Explain why a Phase I algorithm is necessary in this case, and how the modification in Slide 6 achieves dual feasibility.
- Incorrect. The problem is infeasible, not unbounded.
 - Incorrect. The modification is to the objective function, not the constraints.
 - In Slide 5, both the primal and dual problems are infeasible at the origin. The primal problem has negative right-hand sides in its constraints, and the dual problem has a negative right-hand side in one of its constraints. A Phase I algorithm is introduced to find a feasible starting point. Slide 6 modifies the objective function of the primal problem (changing it from $-x_1 + 4x_2$ to $-x_1 - x_2$), which implicitly changes the dual problem, making the origin dual-feasible. This allows the dual simplex method to proceed.
 - Incorrect. The modification is to the objective function, not the constraints, and it aims for dual feasibility first.
 - Incorrect. The problem is infeasible, not unbounded, and the modification is to the primal objective function.
10. Consider the primal problem (P) in Slide 2. The dual simplex method is employed because (P) is not primal feasible. What specific characteristic of the primal problem necessitates the use of the dual simplex method, and why is the dual problem (D) initially feasible?
- The correct answer is A. The presence of negative entries in the right-hand side vector b of the primal problem's constraints indicates that the origin (all variables equal to zero) is not a feasible solution. The dual problem, however, is constructed in such a way that the origin is feasible. This is because the dual constraints are inequalities of the form $A^T y \geq c$, and setting $y = 0$ satisfies these constraints as long as c has non-positive entries (which is often the case when the primal is a maximization problem). The dual simplex method leverages this initial dual feasibility to iteratively move towards primal feasibility and optimality.
 - The primal problem being unbounded does not directly necessitate the dual simplex method. The dual simplex method is used when the primal is infeasible at the origin. The dual problem being a minimization problem is not sufficient to guarantee initial feasibility.
 - Redundant constraints in the primal problem do not automatically imply the need for the dual simplex method. The dual simplex method is specifically designed for cases where the primal is infeasible at the origin.
 - An inconsistent set of constraints in the primal problem would make both the primal and dual problems infeasible. The dual simplex method is used when the primal is infeasible at the origin but the dual is feasible.
 - The dual simplex method applies to linear programs. A non-linear objective function would require different solution techniques.